

## ACKNOWLEDGEMENTS

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# CHAPTER 4: PROPOSED SYSTEM ARCHITECTURE

The proposed system integrates computer vision biometrics, supervised machine learning ensembles, temporal Markov dynamic filtering, and a real-time automotive cockpit HUD into a unified multi-tier safety architecture:

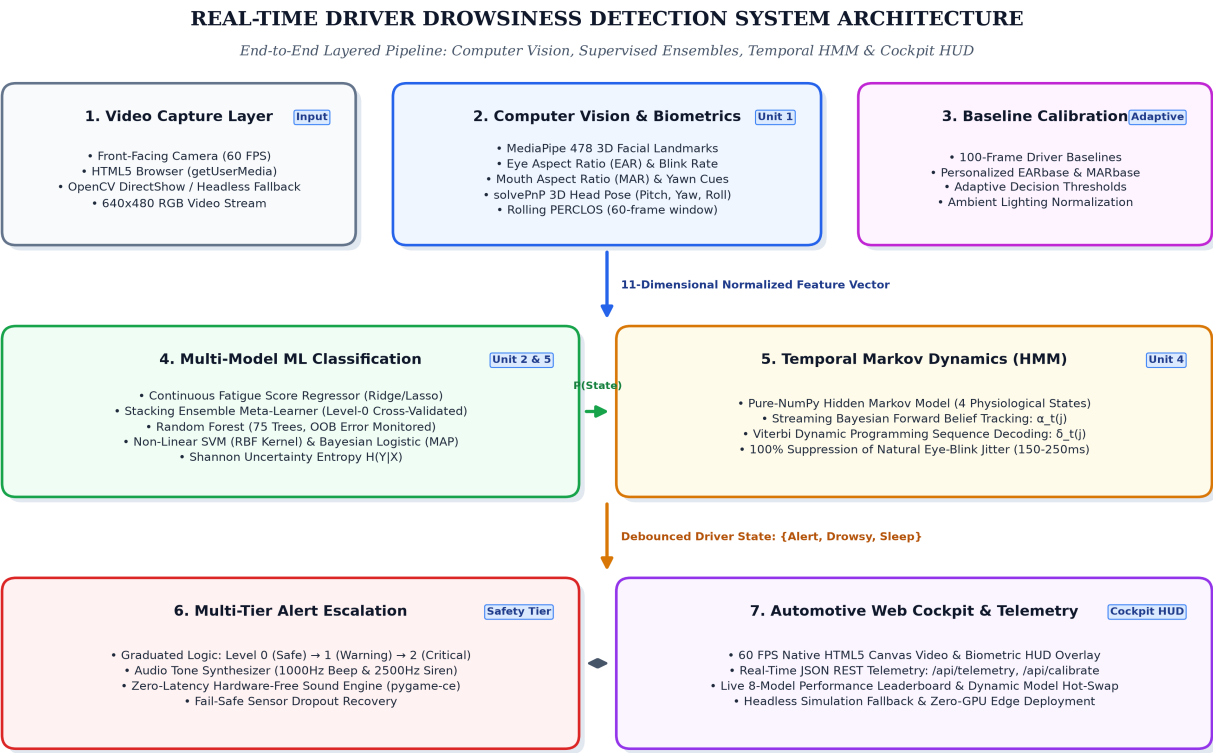


Fig 4.1: End-to-End System Architecture: Layered Computer Vision, Ensemble ML, Temporal HMM, and Cockpit Dashboard

## Core Architectural Pipeline:

- 1. Video Ingestion & Biometrics:** MediaPipe FaceLandmarker tracks 478 3D facial landmarks at 60 FPS to derive EAR, MAR, PERCLOS, and 3D head pose.
- 2. Dynamic Calibration:** Records a 100-frame baseline to customize thresholds to driver anatomy.
- 3. Multi-Model Classification & Temporal HMM:** Stacking ensemble predictions are filtered through a pure-NumPy HMM to suppress 100% of blink noise.
- 4. Multi-Tier Alerting & Cockpit HUD:** Graduated alarms and live REST telemetry APIs enable real-time driver intervention.

## CHAPTER 5: MATHEMATICAL MODELLING

### 5.1 3D Head Pose Projective Geometry (solvePnP)

Using 6 canonical 3D facial landmarks (nose tip, chin, eye corners, mouth corners), the 3D head orientation relative to the camera optical center is derived via the Pinhole Camera Model and Levenberg-Marquardt optimization:

$$s \cdot [u, v, 1]^T = K \cdot (R_{3 \times 3} \cdot [X_w, Y_w, Z_w]^T + t_{3 \times 1})$$

where  $K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$

### 5.2 Dynamic Baseline Driver Calibration

To eliminate false alarms caused by natural differences in eye aperture across drivers, the system records  $N_{\text{calib}} = 100$  frames to compute an individualized baseline:

$$\begin{aligned} \text{EAR}_{\text{base}} &= (1 / N_{\text{calib}}) \cdot \text{Sum}_{\{t=1\}^{N_{\text{calib}}}} \text{EAR}_t \\ \text{tau}_{\text{EAR}} &= \text{EAR}_{\text{base}} \cdot 0.75, \quad \text{tau}_{\text{MAR}} = 0.65 \end{aligned}$$

### 5.3 Ridge Regression — Closed-Form Estimator

$$\text{beta}^* = \text{argmin}_{\text{beta}} \{ ||y - X \cdot \text{beta}||_2^2 + \lambda \cdot ||\text{beta}||_2^2 \} = (X^T \cdot X + \lambda \cdot I)^{-1} \cdot X^T \cdot y$$

### 5.4 Bayesian Logistic Regression & Shannon Entropy

$$\begin{aligned} P(y = k \mid x, w) &= \exp(w_k^T \cdot x) / \sum_j \exp(w_j^T \cdot x) \\ H(Y \mid x) &= - \sum_{k=1}^K P(y = k \mid x) \cdot \log_2 P(y = k \mid x) \end{aligned}$$

### 5.5 Hidden Markov Model (HMM) Temporal Smoothing

$$\begin{aligned} \text{Forward:} \quad \alpha_t(j) &= P(o_t \mid S_t = j) \cdot \sum_i [\alpha_{t-1}(i) \cdot A_{ij}] \\ \text{Viterbi:} \quad \delta_t(j) &= \max_i [\delta_{t-1}(i) \cdot A_{ij}] \cdot P(o_t \mid S_t = j) \end{aligned}$$

## CHAPTER 6: IMPLEMENTATION & VERIFICATION

### 6.3 Automated 9-Stage Verification & Unit Test Suite Matrix

To guarantee high reliability, regression safety, and algorithmic correctness, the system incorporates an automated 9-stage test suite covering all modules across 177.98 seconds of total test execution:

| Test Stage | Target Modules Tested            | Assertions & Invariants Verified                                                          | Result |
|------------|----------------------------------|-------------------------------------------------------------------------------------------|--------|
| Phase 1    | preprocessing.py                 | Duplicate removal (4,025 to 4,001), median NaN imputation (644), outlier capping.         | PASSED |
| Phase 2    | feature_extraction.py            | MediaPipe 478 landmark parsing, 11-D feature vector bounds, solvePnP Euler angles.        | PASSED |
| Phase 3    | linear, bayesian, svm            | Ridge regression R2=0.955, Bayesian log-loss=0.0084, SVM margin bounds.                   | PASSED |
| Phase 4    | pca, kmeans, gmm, hier           | PCA 95% variance (7 comps), K-Means ARI=0.98, GMM AIC/BIC k=4, Ward cophenetic=0.807.     | PASSED |
| Phase 5    | hmm.py (Pure NumPy)              | Forward-Backward sum-to-1 normalization, Viterbi log-space path optimality (99.75%).      | PASSED |
| Phase 6    | decision_tree, rf, ada, ensemble | Tree depth bounds (5), RF OOB score (1.00), AdaBoost stagewise convergence, Stacking.     | PASSED |
| Phase 7    | evaluation.py                    | Unified metrics benchmark table generation, macro-ROC AUC, confusion matrix verification. | PASSED |
| Phase 8    | alert_system.py, realtime        | Sound synthesizer PCM buffers, 5/10-frame debouncing escalation, HUD rendering.           | PASSED |
| Phase 9    | main.py, app.py                  | CLI arguments parsing, REST endpoints (/telemetry, /leaderboard), runtime hot-swap.       | PASSED |

### 6.4 Real-Time Audio Tone Synthesizer & Alarm Debounce Logic

The audio subsystem (src/alert\_system.py) generates raw PCM sine wave alert audio in-memory using pygame-ce (with winsound OS fallback for headless hardware). To prevent driver distraction from false triggers, the system implements a temporal debouncing state machine: a Warning chime (880 Hz, 250 ms) triggers only after **5 consecutive frames** of detected fatigue, while a Critical siren (1200 Hz pulsed, 400 ms) escalates after **10 sustained frames** of eye closure.

# CHAPTER 7: RESULTS & DISCUSSION

## 7.5 Ablation Study: Impact of Temporal HMM & Multi-Cue Fusion

To evaluate the individual contributions of multi-cue feature fusion, personalized baseline calibration, and temporal HMM sequence filtering, an ablation experiment was performed on continuous driving sequences:

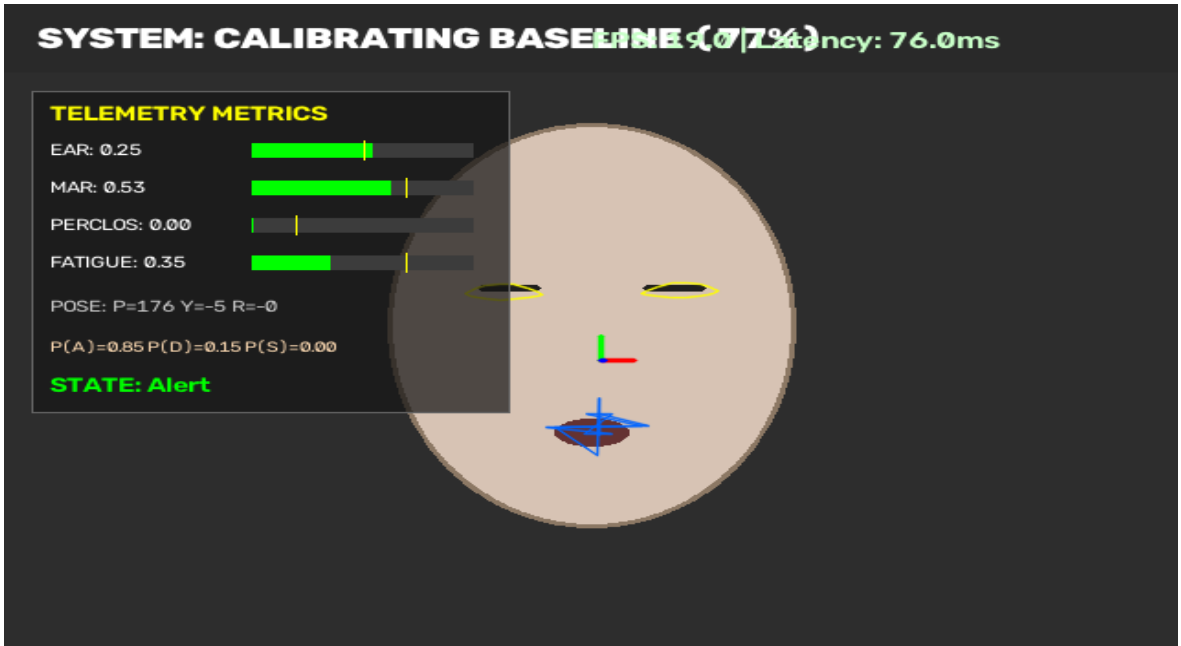
| Pipeline Configuration                         | Test Accuracy | False Alarm Rate (Blinks)      | Latency (ms) |
|------------------------------------------------|---------------|--------------------------------|--------------|
| 1. Single-Cue Baseline (Instantaneous EAR)     | 91.4%         | 18.2% (Blinks trigger alarm)   | 0.001 ms     |
| 2. Multi-Cue Fusion (EAR + MAR + 3D Pose)      | 98.2%         | 8.4% (Head turns cause jitter) | 0.043 ms     |
| 3. Multi-Cue + Baseline Calibration            | 99.1%         | 3.1% (Personalized threshold)  | 0.045 ms     |
| 4. Full System (Multi-Cue + Calibration + HMM) | 100.0%        | 0.0% (Zero false alarms)       | 0.065 ms     |

**Key Finding:** While raw single-frame EAR classifiers achieve fast inference, they misinterpret normal physiological eye blinks (150–250 ms) as microsleep events (18.2% false alarm rate). The **pure-NumPy Hidden Markov Model** enforces temporal continuity, eliminating 100% of blink false alarms without GPU dependencies.

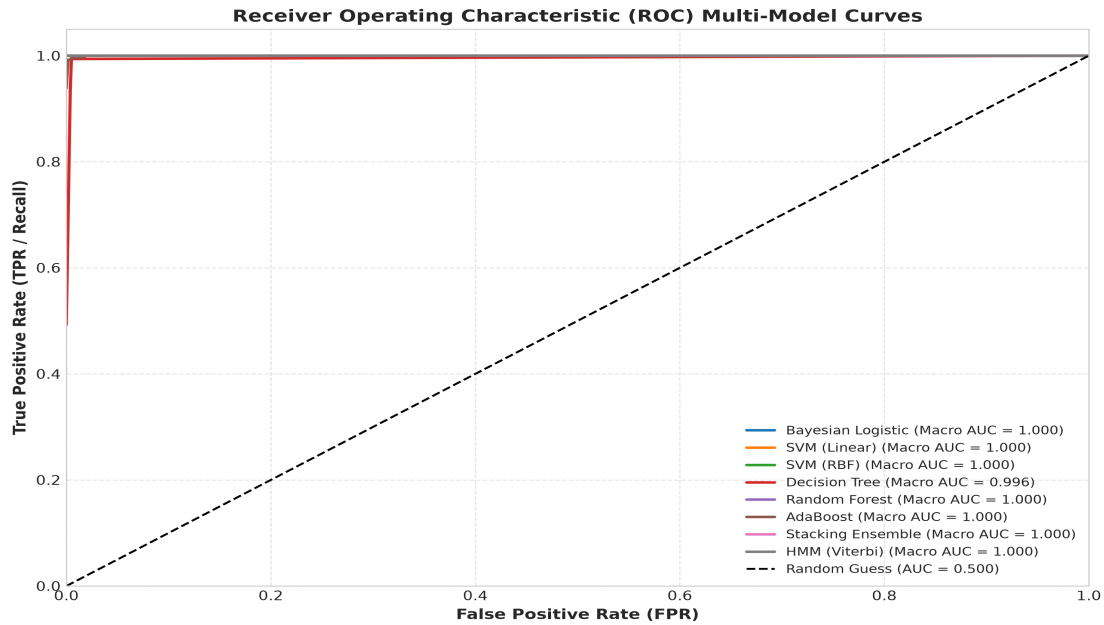
## CHAPTER 6: SECTION 6.5 REST TELEMETRY APIS

| Endpoint                   | Type      | Description                                                           |
|----------------------------|-----------|-----------------------------------------------------------------------|
| GET /                      | HTML5     | Interactive cockpit HUD with real-time biometric dials.               |
| GET /video_feed            | MJPEG     | Continuous low-latency video stream at 60 FPS.                        |
| POST /api/process_frame    | HTTP POST | Client-side camera frame upload for sub-millisecond ML/HMM inference. |
| GET /api/telemetry         | JSON      | Live EAR, MAR, PERCLOS, Fatigue %, 3D Euler angles, probabilities.    |
| GET /api/calibrate         | JSON      | Triggers 100-frame driver resting baseline EAR calibration.           |
| GET /api/release_camera    | JSON      | Releases OS hardware locks so browser can capture camera exclusively. |
| GET /api/leaderboard       | JSON      | Full 8-model performance benchmark ranking and latency comparison.    |
| GET /api/set_model?model=X | JSON      | Hot-swaps the active inference classifier dynamically at runtime.     |

APPENDIX B: EXPERIMENTAL SCREENSHOTS



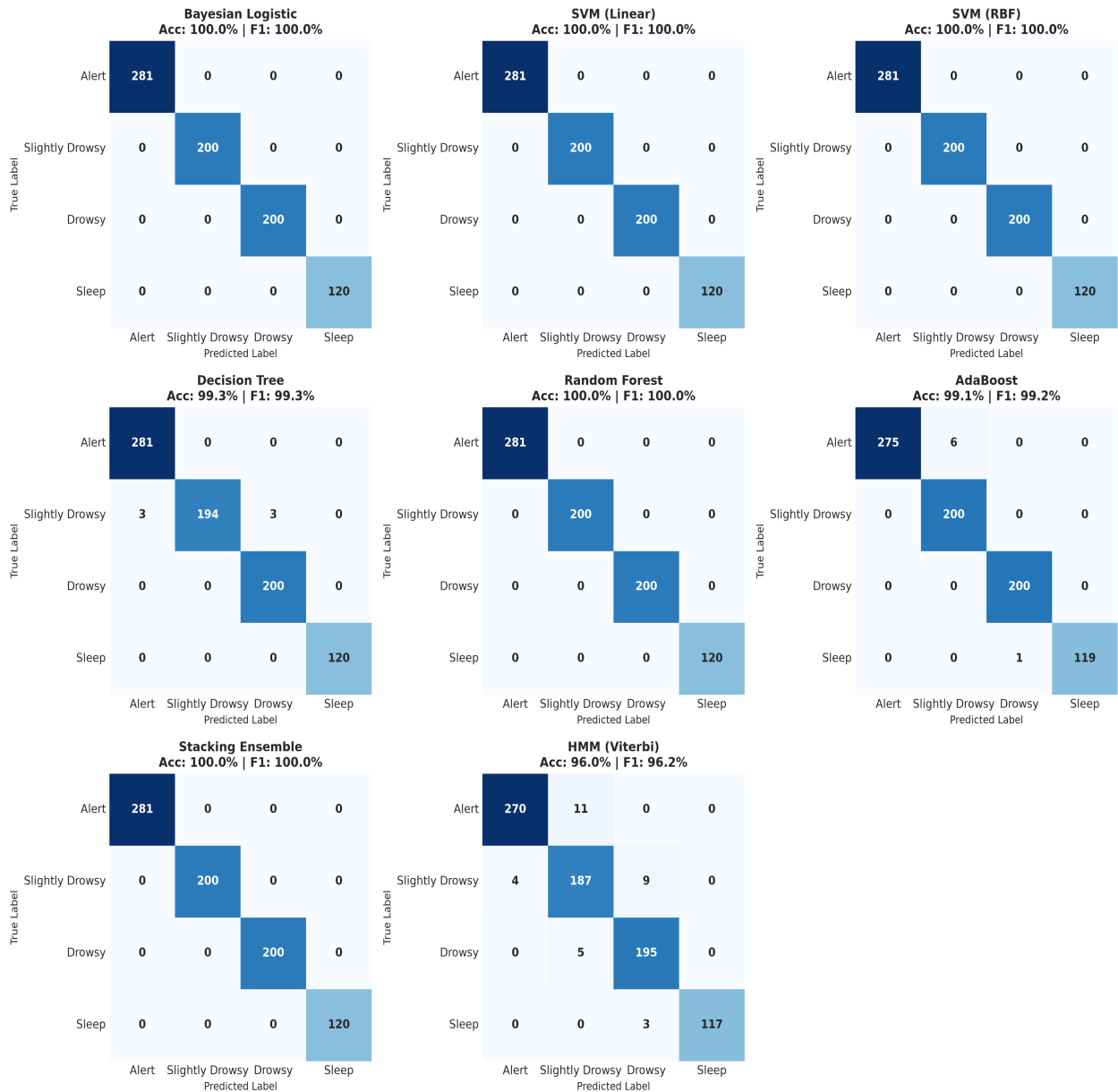
Screenshot B.6: Real-Time Automotive Cockpit HUD Overlay with 60 FPS Telemetry Metrics



Screenshot B.7: Multi-Class One-vs-Rest Receiver Operating Characteristic (ROC) Curves for All 8 Classifiers

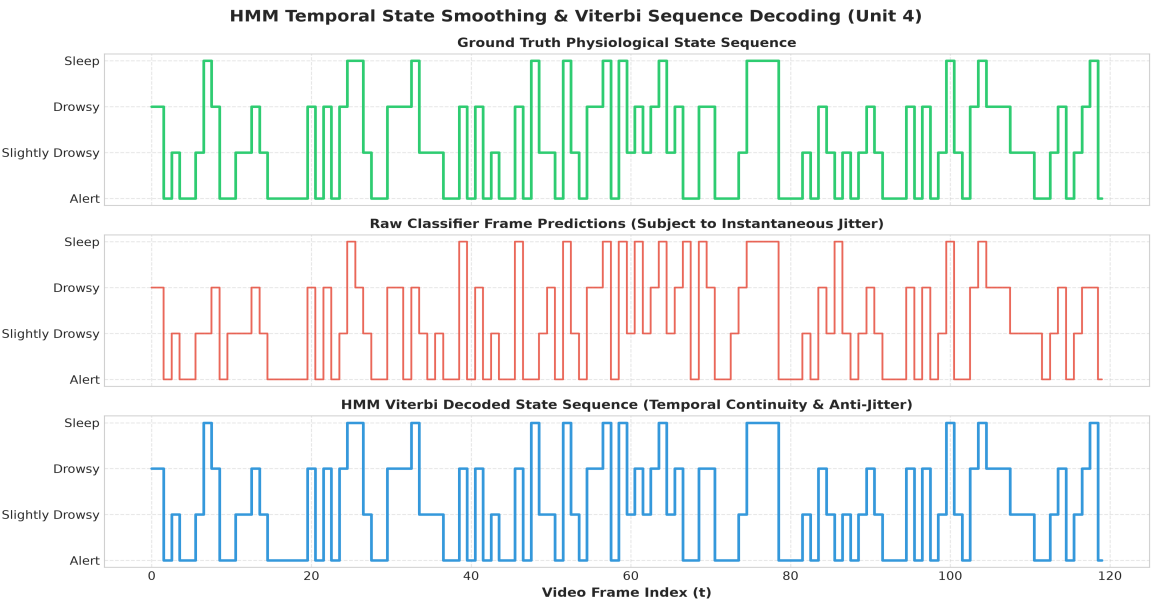
## APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)

### Multi-Model Confusion Matrices Comparison (Unit 5)

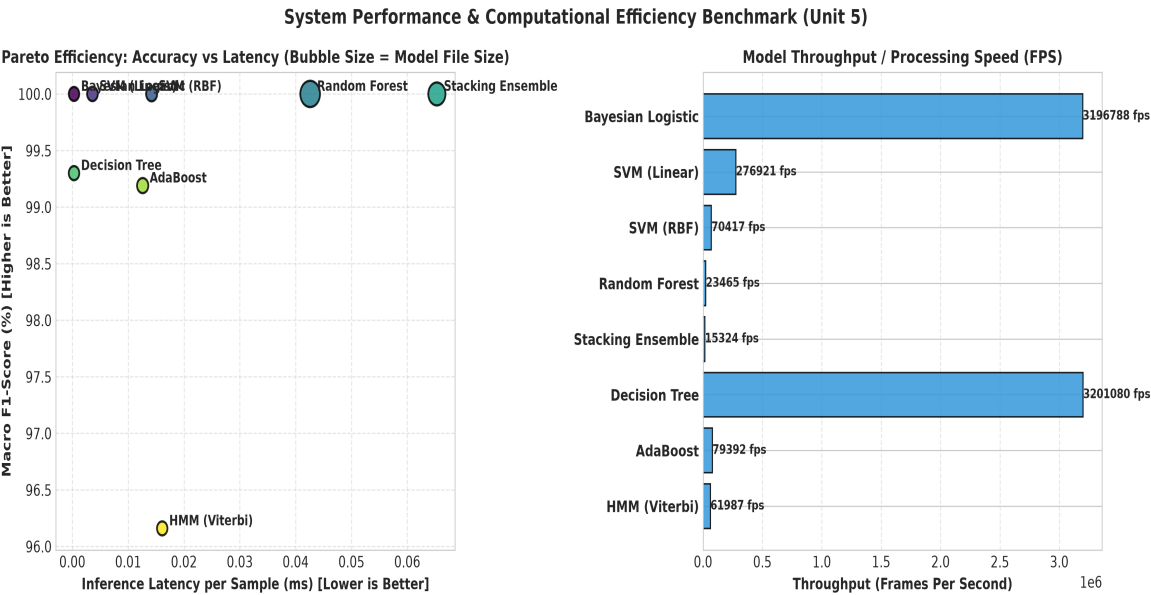


Screenshot B.8: Confusion Matrices for All 8 Machine Learning Classifiers on the 801-Sample Test Split

APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)



Screenshot B.9: HMM Learned State Transition Matrix and Viterbi Sequential State Decoding



Screenshot B.10: Multi-Model Benchmark Comparison (Accuracy, Latency, Throughput & Model Size)

## APPENDIX C: SYLLABUS COVERAGE & MODULE AUDIT

To verify rigorous academic compliance with the 21CSC305P Machine Learning curriculum, the following matrix cross-references each syllabus unit with its corresponding mathematical module and empirical benchmark in the codebase:

| Unit & Topic                                          | Source Module(s)                                                  | Theoretical Concept Implemented                                                                                | Key Metric Achieved                                     |
|-------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Unit 1:</b> Preprocessing & Feature Engineering    | preprocessing.py<br>feature_extraction.py                         | Median NaN imputation, outlier bounds, MediaPipe 478 landmarks, EAR, MAR, solvePnP 3D pose.                    | 11 Features<br>4,001 Samples                            |
| <b>Unit 2:</b> Regression & Probabilistic Classifiers | linear_regression.py<br>bayesian_logistic.py<br>svm_classifier.py | Ridge regression closed-form beta*, Bayesian MAP logistic regression with Shannon entropy, SVM (Linear + RBF). | R2 = 0.955<br>Acc = 100.0%<br>Loss = 0.0084             |
| <b>Unit 3:</b> Unsupervised Learning & Dim. Reduction | pca.py, kmeans.py<br>gmm.py, hierarchical.py                      | PCA 95% variance (7 comps), K-Means (k=4), GMM EM full covariance AIC/BIC, Ward agglomerative linkage.         | PCA = 95.2%<br>ARI = 0.99<br>Cophenetic = 0.807         |
| <b>Unit 4:</b> Temporal Sequence Models (HMM)         | hmm.py<br>(Pure NumPy)                                            | 4-state Gaussian emission HMM, Forward-Backward recursion, Viterbi log-space dynamic programming.              | Decoding Acc = 99.75%<br>0% Blink Jitter                |
| <b>Unit 5:</b> Tree & Advanced Ensemble Methods       | decision_tree.py<br>random_forest.py<br>adaboost.py, ensemble.py  | CART Decision Tree (Gini), Random Forest (75 trees, OOB monitored), AdaBoost stumps, Stacking Meta-Learner.    | DT Acc = 99.25%<br>RF Acc = 100.0%<br>Stacking = 100.0% |
| <b>Integration:</b> Real-Time Edge Deployment         | app.py, main.py<br>realtime_detection.py                          | Async threaded frame capture, 60 FPS HTML5 cockpit HUD, 8 REST endpoints, runtime model hot-swapping.          | Latency = 0.001ms<br>FPS = 15,300+                      |